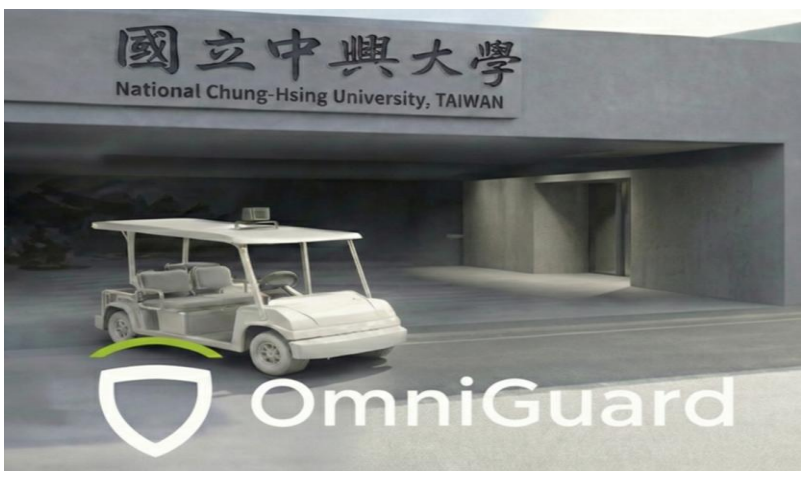
From Simulation to Reality : OmniGuard, the AI Campus Patrol Rover



AI autonomous inspection system

- MSI EdgeXpert on NVIDIA DGX Spark™
- NVIDIA MGX™ Server
- NVIDIA RTX PRO™ 6000 Blackwell Server Edition
- NVIDIA Jetson Thor™
- NVIDIA Omniverse™
- NVIDIA Isaac Sim™

Project Objectives

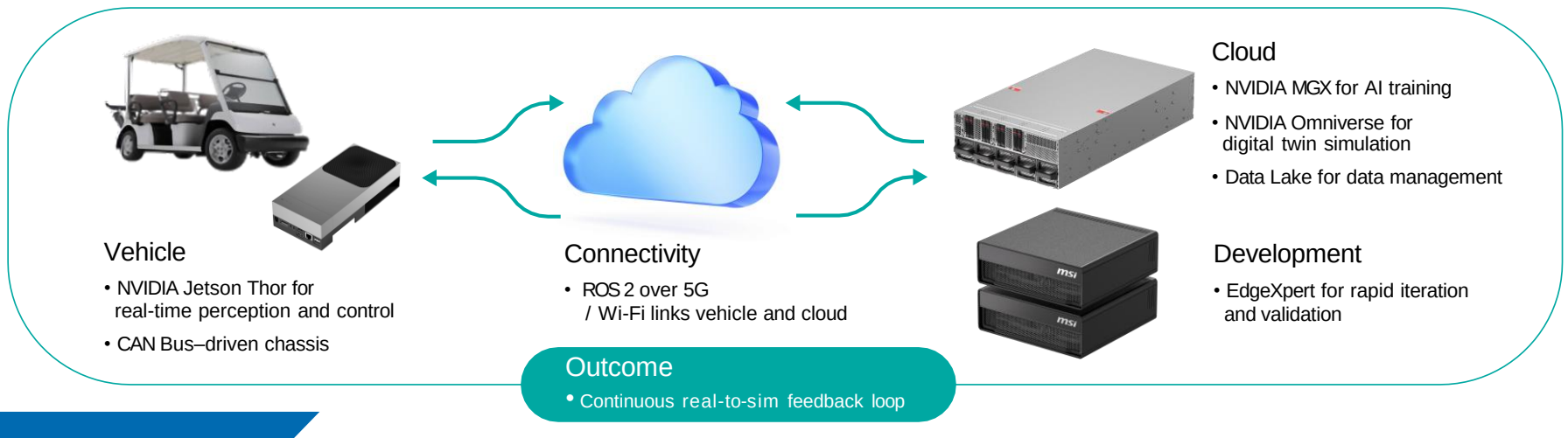
Primary Objectives

1. Build an Autonomous Campus Patrol Vehicle
Enable all-weather, unmanned autonomous patrols across the campus.
2. Create a Digital Twin Environment
Use NVIDIA Omniverse simulation to accelerate AI training and validation.
3. Perform Intelligent Inspection Tasks
Accurately identify potential risks, such as personnel and fire hazards, during both day and night.
4. Adapt to Post-Disaster Scenarios
Automatically detect road damage after typhoons and intelligently replan navigation routes.

Extended Objectives

5. AI Model Self-Learning
Continuously evolve inspection capabilities to improve detection accuracy.
6. Integrate with Campus IoT Platforms
Connect inspection data to the campus-wide IoT management system.

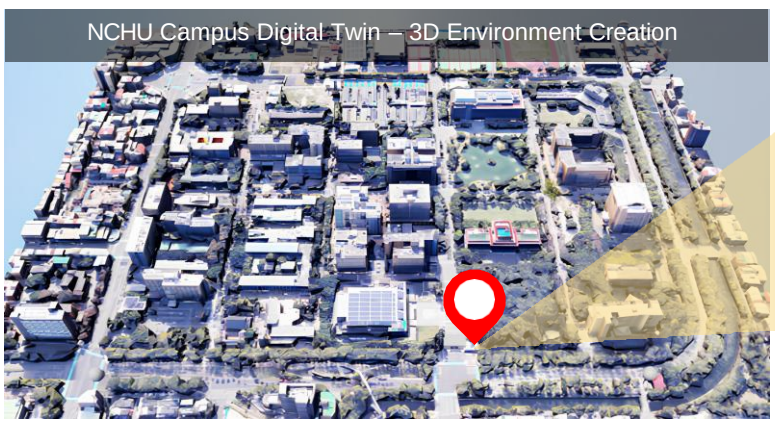
System Architecture



Results

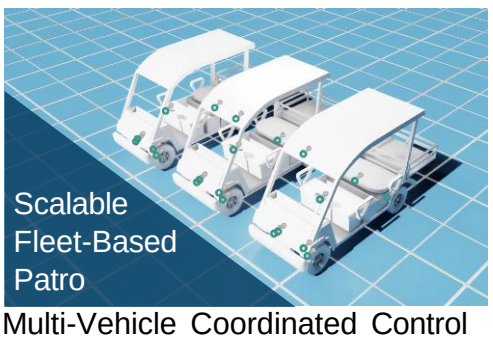
- Demonstrates a scalable AI digital twin platform built on the NVIDIA ecosystem
- Enables reliable Sim-to-Real deployment through high-fidelity simulation and edge execution
- Establishes a robust and extensible development loop for autonomous systems
- Supports cross-vehicle and cross-environment deployment
- Provides a practical foundation for future digital twin applications, including:
  - Smart campuses
  - Industrial sites
  - Large-scale intelligent environments

Campus Digital Twin Simulation Built on NVIDIA Omniverse and Isaac Sim

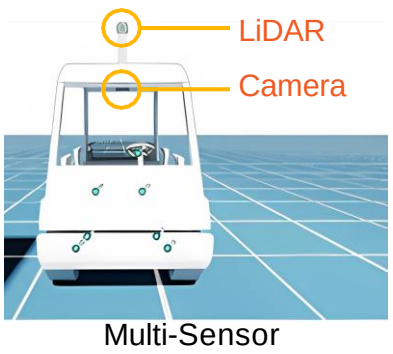


Control Center

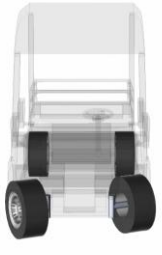
Environmental Validation



Multi-Vehicle Coordinated Control



Multi-Sensor Simulation



Ackermann Steering Model